

Definite Integration

Introduction

$$\int_a^b f(x) dx = (g(x))_a^b = g(b) - g(a)$$

$$\begin{aligned} & \int_0^1 \frac{1}{4x+3} dx \\ &= \frac{1}{4} [\ln |4x+3|]_0^1 \\ &= \frac{1}{4} [\ln 7 - \ln 3] \end{aligned}$$

$$\begin{aligned}
 \int_{-1}^0 2^x dx &= \left[\frac{2^x}{\ln 2} \right]_{-1}^0 \\
 &= \frac{1}{\ln 2} [1 - 2^{-1}] \\
 &= \frac{1}{\ln 2} \left[1 - \frac{1}{2} \right] \\
 &= \frac{1}{2 \ln 2}
 \end{aligned}$$

Definite Integration by Parts

$$\int_0^{\pi} \sin x \, dx = 2$$

$$\int_0^{2\pi} \sin x \, dx = 0$$

$$\int_0^{\pi} \cos x \, dx = 0$$

$$\int_0^{\pi/2} \cos x \, dx = 1$$

Definite Integration by Parts

$$\begin{aligned} \int_0^1 x e^x \, dx &= (x e^x)_0^1 - \int_0^1 1 \cdot e^x \, dx \\ &= (x e^x)_0^1 - (e^x)_0^1 \\ &= (e - 0) - (e - 1) \\ &= 1 \end{aligned}$$

Properties of Definite Integration

$$\int_a^b f(x) dx = \int_a^b f(t) dt$$

Proof:

$$\text{Let } x = t$$

$$dx = dt$$

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

Substitution in Definite Integration

$$I_1 = \int_e^{e^2} \frac{1}{\ln x} dx$$

$$\text{Let } \ln x = t \implies x = e^t$$

$$dx = e^t dt$$

$$I_1 = \int_1^2 \frac{e^t}{t} dt$$

$$I_2 = \int_1^2 \frac{e^x}{x} dx$$

$$I_1 = I_2$$

**Limits also change with
Substitution :**

Substitution used : $\ln x = t$

Original Lower Limit = e

Calculations : $\ln e = 1$

New Lower Limit = 1

Original Upper Limit = e^2

Calculations : $\ln e^2 = 2$

New Upper Limit = 2

Properties of Definite Integration

$$\begin{aligned}\int_a^b f(a + b - x) \, dx &= - \int_b^a f(t) \, dt \\ &= \int_a^b f(t) \, dt = \int_a^b f(x) \, dx\end{aligned}$$

$$(\text{Let } a + b - x = t \rightarrow -dx = dt)$$

Limits also change with Substitution :

Substitution used : $t = a + b - x$

Original Lower Limit : $x = a$

Calculations : $t = a + b - (a) = b$

New Lower Limit : $t = b$

Original Upper Limit : $x = b$

Calculations : $t = a + b - (b) = a$

New Upper Limit : $t = a$

Evaluate: $\int_0^1 x(1-x)^{99} dx$

$$I = \int_0^1 x(1-x)^{99} dx \quad \dots (1)$$

$$(x \rightarrow 0 + 1 - x)$$

$$I = \int_0^1 (1-x)x^{99} dx \quad \dots (2)$$

$$I = \int_0^1 (x^{99} - x^{100}) dx$$

$$= \left(\frac{x^{100}}{100} - \frac{x^{101}}{101} \right)_0^1$$

$$= \frac{1}{100} - \frac{1}{101}$$

$$= \frac{1}{10100}$$

$$\int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(x) \text{ is even} \\ 0, & \text{if } f(x) \text{ is odd} \end{cases}$$

Even function: $f(-x) = f(x)$

Odd function: $f(-x) = -f(x)$

Evaluate $\int_{-\pi/2}^{\pi/2} \sin x dx$

Since $\sin x$ is an odd function:

$$I = \int_{-\pi/2}^{\pi/2} \sin x dx = 0$$

Alternative calculation:

$$I = -[\cos x]_{-\pi/2}^{\pi/2}$$

$$\begin{aligned} I &= -(\cos(\pi/2) - \cos(-\pi/2)) \\ &= -(0 - 0) = 0 \end{aligned}$$

$$\int_0^{2a} f(x) dx = \int_0^a [f(x) + f(2a - x)] dx$$

$$= \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(x) = f(2a - x) \\ 0, & \text{if } f(x) = -f(2a - x) \end{cases}$$

Evaluate $I = \int_0^{2\pi} \cos^3 x dx$

$$(x \rightarrow 0 + 2\pi - x)$$

$$f(x) = \cos^3 x$$

$$f(2\pi - x) = (\cos(2\pi - x))^3 = (\cos x)^3 = f(x)$$

$$I = 2 \int_0^{\pi} (\cos^3 x) dx$$

$$(x \rightarrow 0 + \pi - x)$$

$$f(\pi - x) = (\cos(\pi - x))^3 = (-\cos x)^3 = -f(x)$$

$$I = 2 \int_0^{\pi} (\cos^3 x) dx = 0$$

If $f(x)$ is a periodic function with period T , then $\rightarrow$

$$\int_0^{nT} f(x) dx = n \int_0^T f(x) dx, \quad n \in \mathbb{Z}$$

Evaluate $\int_0^{1000\pi} \sqrt{1 + \cos 2x} \, dx$

$$I = \int_0^{1000\pi} \sqrt{2 \cos^2 x} \, dx$$

$$= \sqrt{2} \int_0^{1000\pi} |\cos x| \, dx$$

$$= \sqrt{2} \cdot 1000 \int_0^{\pi} |\cos x| \, dx$$

$$= \sqrt{2} \cdot 1000 \left[\int_0^{\pi/2} \cos x \, dx + \int_{\pi/2}^{\pi} (-\cos x) \, dx \right]$$

$$= 1000\sqrt{2} \left[(\sin x)_0^{\pi/2} - (\sin x)_{\pi/2}^{\pi} \right]$$

$$= 1000\sqrt{2}[1 - (0 - 1)] = 2000\sqrt{2}$$

If $f(x)$ is a periodic function
with period T , then $\rightarrow$

$$\int_{a+nT}^{b+nT} f(x) dx = \int_a^b f(x) dx$$

$$\int_{nT}^{a+nT} f(x) dx = \int_0^a f(x) dx$$

$$\int_a^{a+nT} f(x) dx = n \int_0^T f(x) dx$$

$$\int_{mT}^{nT} f(x) dx = (n - m) \int_0^T f(x) dx$$

Substitution in Definite Integration

$$\text{If } F(x) = \int_{g(x)}^{h(x)} f(t) dt, \text{ then } \rightarrow$$

$$\frac{d}{dx} F(x) = f(h(x)) \cdot h'(x) - f(g(x)) \cdot g'(x)$$

$$\text{Evaluate } f'(x) \text{ for } f(x) = \int_x^{x^2} \cos t dt$$

$$f'(x) = (\cos x^2)2x - (\cos x)(1)$$

Reduction Formulae

$$I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$I_n = \left(-\frac{\cos x \sin^{n-1} x}{n} + \frac{(n-1)}{n} I_{n-2} \right)$$

$$I_n = \left[-\frac{\cos x \sin^{n-1} x}{n} \right]_0^{\frac{\pi}{2}} + \frac{(n-1)}{n} I_{n-2}$$

$$I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n-1)}{n} I_{n-2}$$

$$\text{Let } I_6 = \int_0^{\frac{\pi}{2}} \sin^6 x \, dx$$

$$I_6 = \frac{5}{6} I_4$$

$$I_6 = \frac{5}{6} \left[\frac{3}{4} I_2 \right]$$

$$= \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} I_0 = \frac{15}{48} \times \frac{\pi}{2} = \frac{15\pi}{96}$$

$$I_0 = \int_0^{\frac{\pi}{2}} 1 \, dx = \frac{\pi}{2}$$

$$I_1 = \int_0^{\frac{\pi}{2}} \sin x \, dx = -(\cos x) \Big|_0^{\frac{\pi}{2}} = 1$$

$$I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$$

$$I_n = \left(\frac{\sin x \cos^{n-1} x}{n-1} + \frac{n-1}{n} I_{n-2} \right) \Big|_0^{\frac{\pi}{2}}$$

$$I_n = \left(\frac{n-1}{n} \right) I_{n-2}$$

$$I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx = \left(\frac{n-1}{n} \right) I_{n-2}$$

$$I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$$

$$\text{If } I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$$

$$I_n + I_{n-2} = \frac{1}{(n-1)}$$

$$I_5 + I_3 = \int_0^{\frac{\pi}{4}} (\tan^5 x + \tan^3 x) \, dx = \frac{1}{4}$$

$$I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$$

Reduction formula: $I_n + I_{n-2} = \frac{1}{n-1}$

First, find I_0 and I_2 :

$$I_0 = \int_0^{\frac{\pi}{4}} 1 \, dx = [x]_0^{\frac{\pi}{4}} = \frac{\pi}{4}$$

$$I_2 + I_0 = \frac{1}{2-1} = 1$$

$$I_2 = 1 - I_0 = 1 - \frac{\pi}{4}$$

Next, find I_4 :

$$I_4 + I_2 = \frac{1}{4-1} = \frac{1}{3}$$

$$\begin{aligned} I_4 &= \frac{1}{3} - I_2 = \frac{1}{3} - \left(1 - \frac{\pi}{4}\right) \\ &= \frac{1}{3} - 1 + \frac{\pi}{4} = \frac{\pi}{4} - \frac{2}{3} \end{aligned}$$

Finally, find I_6 :

$$I_6 + I_4 = \frac{1}{6-1} = \frac{1}{5}$$

$$I_6 = \frac{1}{5} - I_4 = \frac{1}{5} - \left(\frac{\pi}{4} - \frac{2}{3} \right)$$

$$I_6 = \frac{1}{5} - \frac{\pi}{4} + \frac{2}{3}$$

Combining the fractional terms:

$$I_6 = \frac{3+10}{15} - \frac{\pi}{4} = \frac{13}{15} - \frac{\pi}{4}$$

The value of $\int_0^{\frac{\pi}{4}} \tan^6 x \, dx$ is $\frac{13}{15} - \frac{\pi}{4}$

Walli's Formula

$$\text{If } I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x \, dx$$

$$I_{m,n} = \frac{(m-1)(m-3)(m-5)\cdots(n-1)(n-3)(n-5)\cdots}{(m+n)(m+n-2)(m+n-4)\cdots} \cdot \frac{\pi}{2}$$

$\implies$ (when both, m and n are even)

$$I_{m,n} = \frac{(m-1)(m-3)(m-5)\cdots(n-1)(n-3)(n-5)\cdots}{(m+n)(m+n-2)(m+n-4)\cdots} \cdot 1$$

$\implies$ (otherwise)

Question:

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cos^2 x (\sin x + \cos x) dx$$

Solution:

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^3 x \cos^2 x dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cos^3 x dx$$

$$I = 0 + 2 \int_0^{\frac{\pi}{2}} \sin^2 x \cos^3 x dx$$

Using Walli's Formula:

$$I = 2 \times \frac{(2-1) \times (3-1)}{(2+3) \times (2+3-2) \times (2+3-4)}$$

$$I = 2 \times \frac{1 \times 2}{5 \times 3 \times 1}$$

$$I = \frac{4}{15}$$